Fowards

- In this lecture, we begin our study of derivatives
- We will start with *forwards*
- Standardized implementations of forwards are called *futures*)
- Henceforth, I will refer to everything as a "forward."
- We will then use forwards to study
  (1) swaps
  (2) exchange rates
- Common examples: foreign currencies, agricultural products, metals and energy, interest rate futures, equity indexes

Vocabulary

- *Underlying*
- A *counter-party*
- *Cash settlement* means delivering the cash value of the underlying asset (rather than the asset itself) to satisfy the contract
- A *forward* is a contract that requires future delivery of asset at agreed-upon price
- The *future price* is the agreed-upon price to be paid on a futures contract at maturity
- The *spot price* is the actual market price of the commodity at the time of the delivery.

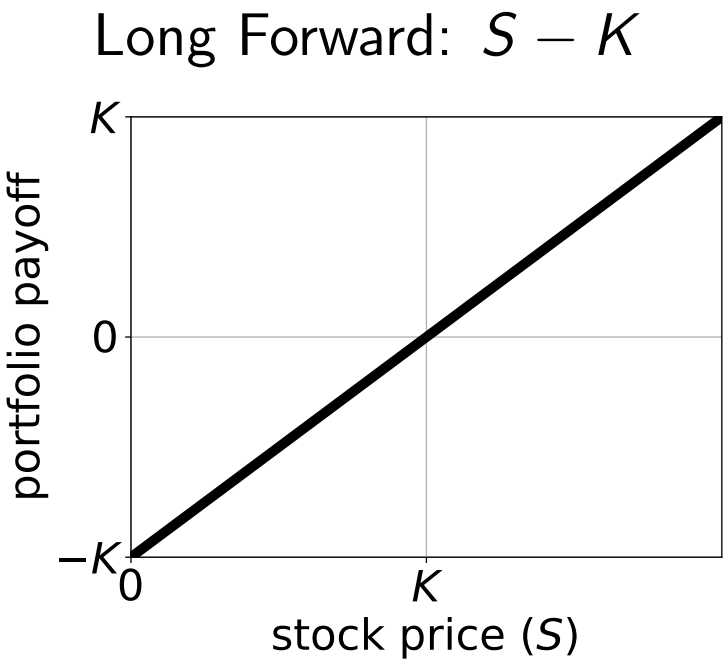
Trading Strategies

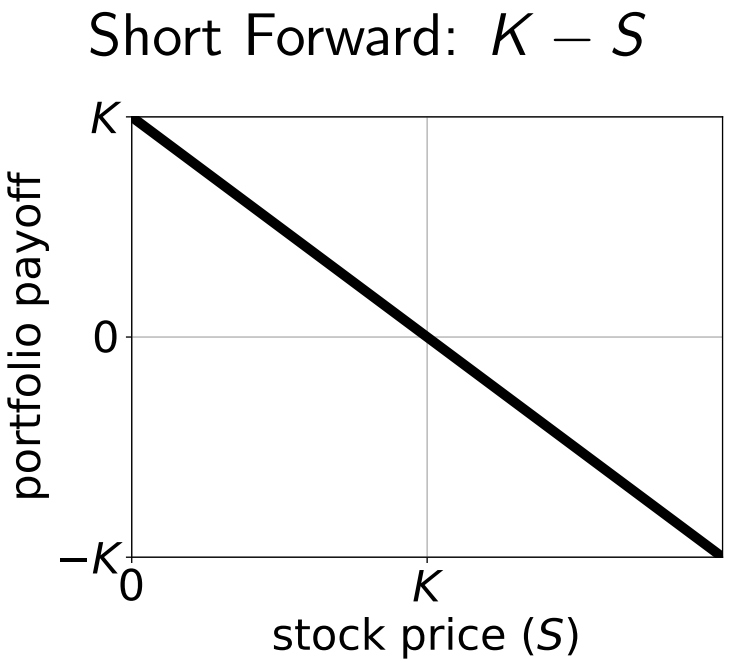
- Long position: commit to *purchase* asset
- Short position: commit to *deliver* asset
- Positions settled at maturity (no initial cash flow)
- Speculation
  - Short if you believe price will fall
  - Long if you believe price will rise
- Hedging
  - Long: Endowment fund will purchase stock in 3 months; manager buys forwards now to protect against rise in price
  - Short: Hedge fund invests in long-term bonds; manager worries interest rates may increase and sells forwards

Forwards Contracts

- The trader in the long position is said to "buy" a contract; the short-side trader "sells" a contract.
- Basis: difference between futures price and spot price $K - S_t$
- In practice, traders may be required to post margin.
- Because <u>both</u> parties to a futures contract are exposed to losses, <u>both</u> must post margins.

Long and Short Forward





Corporate Applications

- Consider the case of an oil production company. How much will the price of oil be in the future, when they are ready to sell the oil they produced? The revenue of this company depends on oil prices, which are extremely volatile!
- On the other hand, consider a company that has to buy oil as one of the inputs in the production process (airline company). This company has a problem that mirrors that of the oil production company.
- **Can forwards help these two firms?**

Corporate Applications

- **Yes!** The two parties can enter into an agreement today that specifies the price at which the oil producer sells the oil to the other company in the future.
- Such an agreement will reduce the source of risk for both parties! The oil producer delivers oil at a price agreed upon now, regardless of the market price at the time when oil is produced.
- This requires that BOTH parties be willing to lock in the price to be paid or received for delivery of the commodity.
- This can be achieved using forwards.

Corporate Applications

Southwest Airlines gains advantage by hedging on long-term oil contracts

By **JEFF BAILEY** NOV. 28, 2007

CHICAGO — Southwest Airlines, in danger for much of this year of losing its quirky dominance over the U.S. domestic airline industry, could soon be standing, once again, head-and-shoulders above its competition.

Better service? Happier and more productive workers?

Not this time. The reason for Southwest's rapidly increasing advantage over other big airlines is much simpler: It loaded up years ago on hedges against higher fuel prices. And with oil trading above $90 a barrel, most of the rest of the airline industry is facing a huge run-up in costs, and Southwest is not.

Southwest owns long-term contracts to buy most of its fuel at the equivalent of $51-a-barrel oil through 2009. The value of those hedges soared as oil raced above the $90-a-barrel mark and they are now worth more than $2 billion.

Those gains would mostly be realized during the next two years.

Other major airlines passed on buying all but the shortest-term insurance against high fuel prices, giving Southwest executives a mild case of schadenfreude.

"It's true," said Scott Topping, treasurer at Southwest and keeper of the hedges. "We're not sure what to root for," in terms of oil prices. Southwest is hurt, too, by higher fuel prices, but far less than competitors, giving the carrier a distinct

Pricing

- Spot-Futures Parity-Theorem $K \overset{\text{“should”}}{=} S_0(1+r_F)^T$

Security	Q	CF_0	CF_T
Forward	1	0	$S_T - K$
Stock	1	$-S_0$	S_T
Risk-Free	$-K/(1+r)^T$	$K/(1+r)^T$	$-K$
		$-S_0 + K/(1+r)^T$	$S_T - K$

Problem 1

You're in sales at Goldman. A client wants a five-year forward on \$1m of the S&P 500. The risk-free rate is 10%. What should the price K be? How do you hedge the position? What if the client tells you that he is willing to buy the forward for \$2m. What is your profit?

Problem 2

Suppose that you buy a one-year forward on 10,000 shares of SPY, which currently sells for \$275. The risk-free rate is 10%. You are required to post an initial margin of 20% and meet a maintenance margin of 10%

- What should the forward price be?
- How much margin (\$) must you post?
- If SPY drops by 5% tomorrow, what will the new forward price be?
- How much margin (%) do you have left?
- Will you receive a margin call?
- What is your return?

Pricing Forwards

- Suppose you go long a forward that expires in <u>two</u> years.
- Step forward one year.
- Consider a new forward that expires in <u>one</u> year.
- How much could you sell your old forward for (say f_0)?

Pricing Forwards

How do you price anything? Build a replicating portfolio!

Security	Q	CF_1	CF_2
Old Forward (K_0)	$+1$	$-f_0$	$S_2 - K_0$
	Replicating Portfolio		
New Forward (K_1)	$+1$	0	$S_2 - K_1$
Risk Free	$PV(K_1 - K_0)$	$-PV(K_1 - K_0)$	$K_1 - K_0$
		$-PV(K_1 - K_0)$	$S_2 - K_0$

$$\boxed{f_0 \overset{\text{“should”}}{=} PV(K_1 - K_0)} \tag{1}$$

Pricing Forwards: Example

- $S_0 = \$100$, $S_1 = \$115$, $r = 10\%$
- The old and new forward prices are

$$K_0 = S_0(1+r)^2 = \$121 \tag{2}$$
$$K_1 = S_1(1+r) = \$126 \tag{3}$$

- The old forward trades for

$$f_0 = \frac{K_1 - K_0}{1+r} = \frac{\$126 - \$121}{1+.1} = \$4.54 \tag{4}$$

Pricing Forwards: Example

- $S_0 = \$100$, $S_1 = \$110$, $r = 10\%$
- The old and new forward prices are

$$K_0 = S_0(1+r)^2 = \$121 \tag{5}$$
$$K_1 = S_1(1+r) = \$121 \tag{6}$$

- The old forward trades for

$$f_0 = \frac{K_1 - K_0}{1+r} = \frac{\$121 - \$121}{1+.1} = \$0 \tag{7}$$

Application I: Swaps

- A swap is a contract in which parties "swap" cash-flows.
- Party A has cash flows $c_1^A, c_2^A, \ldots$
- Party B has cash flows $c_1^B, c_2^B, \ldots$
- A and B swap: B pays A $c_1^B, c_2^B, \ldots$ and A pays B $c_1^A, c_2^A, \ldots$
- Swaps are technically portfolios of forwards

IBM and the World Bank

- Salomon Brothers wrote the first swap in 1981
- The World Bank is an international organization that borrows from developed countries and lends to developing countries
- In 1981, Swiss debt was cheap, so the World Bank wanted to borrow there, but...
- Switzerland limited how much the World Bank could borrow
- IBM had already borrowed in Switzerland and wanted dollars
- Salomon proposed "swapping" cash flows
- The World Bank borrowed dollars
- IBM paid the World Bank's dollar obligations
- The World Bank paid IBM's franc obligations
- It was as if the World Bank borrowed in francs and IBM borrowed in dollars

Interest Rate Swaps

- The Solomon swap was a *currency* swap
- We will start with *interest rate* swaps
- A vanilla swap is "fixed-for-floating"
- The party who is long the swap receives fixed rate cash flows and pays floating rate cash flows

Interest Rate Swaps

Term	Year 0	Year 1
One-Year	${}_0y_1$	${}_1y_1$
Two-Year	${}_0y_2$	$\times$

- Suppose we have the above yield curve
- All bonds on the yield curve are 1k, zero-coupon
- There are two kinds of loans, fixed and floating
- Fixed loan: borrow with <u>known</u> coupons ${}_0c_2 \equiv {}_0y_2 \cdot 1\text{k}$ in years one and two
- Floating loan: borrow with <u>known</u> coupon ${}_0c_1 \equiv {}_0y_1 \cdot 1\text{k}$ in the year one and <u>unknown</u> coupon ${}_1c_1 \equiv {}_1y_1 \cdot 1\text{k}$ in year two

Interest Rate Swaps

- Consider two firms, A and B
- A makes fixed payments on its debt
- B makes floating payments on its debt

Firm A

0	1	2
p_A	$-{}_0c_2$	$-(1 + {}_0c_2)$

Firm B

0	1	2
p_B	$-{}_0c_1$	$-(1 + {}_1c_1)$

Interest Rate Swaps

- If we assume that the original debts were priced correctly,

$$p_A = p_B = 1 \tag{8}$$

- Coupon rate always equals yield, so bond always worth face
- You can show this with a no-arbitrage argument

Interest Rate Swaps

- A <u>makes</u> fixed payments and <u>wants</u> floating payments
- B <u>makes</u> floating payments and <u>wants</u> fixed payments
- Solution: A sells a swap to B

Interest Rate Swaps

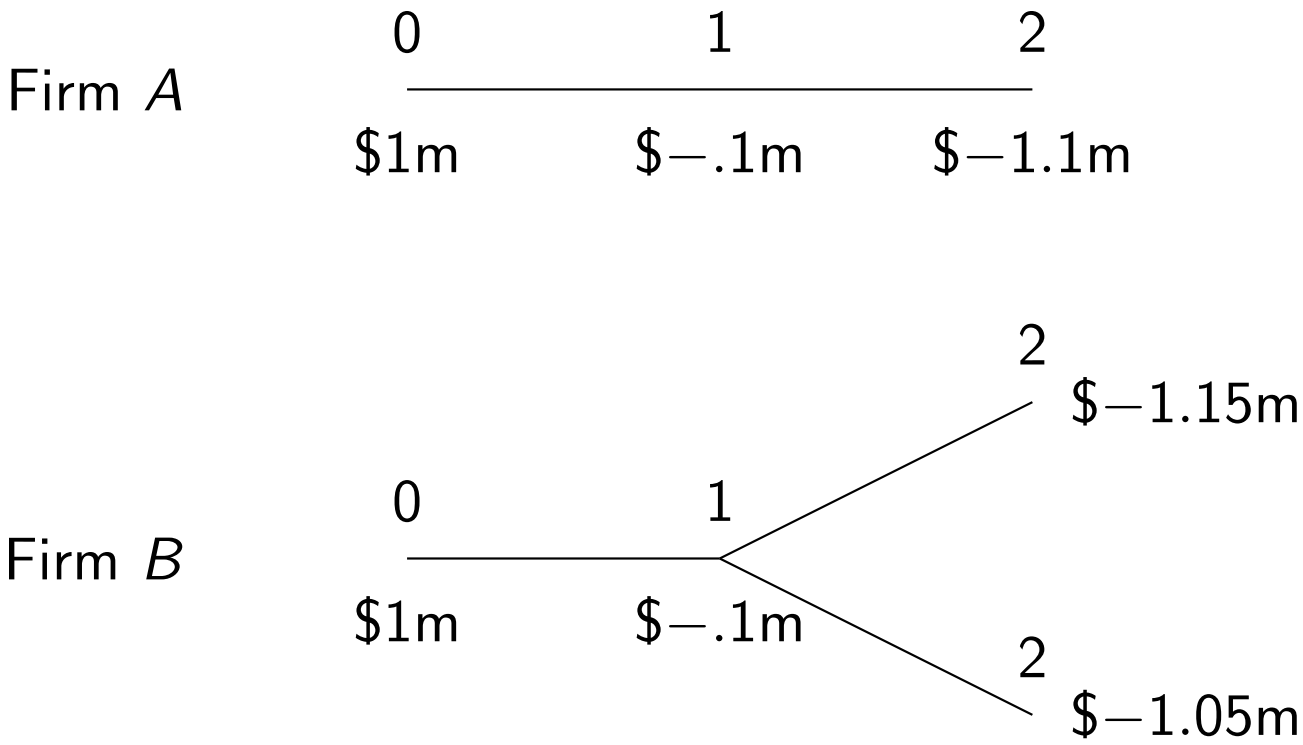
W_0 is the price of the swap

Security	Q	CF_0	CF_1	CF_2
		Firm A		
Loan (Fixed)	$+1$	$+p_A$	$-\,{}_0c_2$	$-(1+\,{}_0c_2)$
Swap	$+1$	$-w_0$	${}_0c_2-\,{}_0c_1$	${}_0c_2-\,{}_1c_1$
		p_A-w_0	$-\,{}_0c_1$	$-(1+\,{}_1c_1)$
		Firm B		
Loan (Floating)	$+1$	$+p_B$	$-\,{}_0c_1$	$-(1+\,{}_1c_1)$
Swap	-1	$+w_0$	$-({}_0c_2-\,{}_0c_1)$	$-({}_0c_2-\,{}_1c_1)$
		p_B+w_0	$-\,{}_0c_2$	$-(1+\,{}_0c_2)$

A's payments are now floating and B's payments are now fixed

Interest Rate Swaps: Example

- The rate on two-year debt is 10% this year
- The rate on one-year debt is 10% this year, but the rate on one-year debt could either be 5% or 15% next year.



Interest Rate Swaps: Example

Suppose that $w_0 = 0$. Units are millions of USD.

Security	Q	CF_0	CF_1	$CF_{2,u}$	$CF_{2,d}$
		Firm A			
Loan (Fixed)	$+1$	$+1$	$-.1$	-1.1	-1.1
Swap	$+1$	0	0	$-.05$	$+.05$
		$+1$	$-.1$	-1.15	-1.05
		Firm B			
Loan (Floating)	$+1$	$+1$	$-.1$	-1.15	-1.05
Swap	-1	0	0	$+.05$	$-.05$
		$+1$	$-.1$	-1.1	-1.1

A's payments are now floating and B's payments are now fixed

Interest Rate Swaps: Pricing

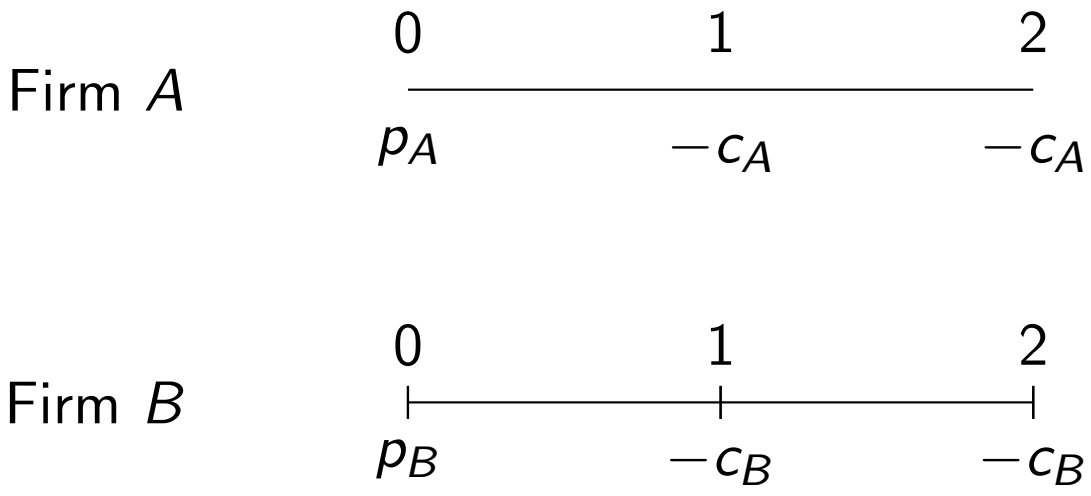
w_0 is the price of the swap

Security	Q	CF_0	CF_1	CF_2
Swap	$+1$	$-w_0$	${}_0c_2 - {}_0c_1$	${}_0c_2 - {}_1c_1$
		Replicating Portfolio		
${}_0c_2$	$+1$	-1	${}_0c_2$	$1 + {}_0c_2$
${}_0c_1$	-1	$+1$	$-(1 + {}_0c_1)$	$\times$
${}_1c_1$	-1	$\times$	$+1$	$-(1 + {}_1c_1)$
		0	${}_0c_2 - {}_0c_1$	${}_0c_2 - {}_1c_1$

Absent arbitrage, $w_0 = 0$.

Another Swap Example

- A borrows p_A, pays coupon c_A each period
- B borrows p_B, pays coupon c_B each period



Another Swap Example

- If we assume that the original debts were priced correctly,

$$p_A = \frac{c_A}{1+y} + \frac{c_A}{(1+y)^2} \tag{9}$$

$$p_B = \frac{c_B}{1+y} + \frac{c_B}{(1+y)^2} \tag{10}$$

Interest Rate Swaps

w_0 is the price of the swap

Security	Q	CF_0	CF_1	CF_2
		Firm A		
Loan	$+1$	$+p_A$	$-c_A$	$-c_A$
Swap	$+1$	$-w_0$	$c_A - c_B$	$c_A - c_B$
		$p_A - w_0$	$-c_B$	$-c_B$
		Firm B		
Loan	$+1$	$+p_B$	$-c_B$	$-c_B$
Swap	-1	$+w_0$	$-(c_A - c_B)$	$-(c_A - c_B)$
		$p_B + w_0$	$-c_A$	$-c_A$

Interest Rate Swaps: Pricing

w_0 is the price of the swap

Security	Q	CF_0	CF_1	CF_2
Swap	$+1$	$-w_0$	$c_A - c_B$	$c_A - c_B$
Replicating Portfolio				
Loan A	$+1$	$-p_A$	$+c_A$	$+c_A$
Loan B	-1	$+p_B$	$-c_B$	$-c_B$
		$-p_A + p_B$	$c_A - c_B$	$c_A - c_B$

$$\boxed{w_0 \overset{\text{“should”}}{=} p_A - p_B} \tag{11}$$

Application II: Exchange Rates

- International finance is a large and rich topic
- Today, we're going to price currencies
- We're going to study a couple of practical examples

Currencies

- Currencies are like tacos
- I go to 3 Amigos and buy tacos, I go to Chase and buy pesos
- In both cases, I pay with dollars
- Let E denote the price of foreign currency
- E.g. a peso sells for $E = \$.05$

Example 1

- Two countries, domestic (D) and foreign (F)
- r_D is the risk-free rate in D, r_F in F
- E_0 is the spot price of F's currency
- K_0 is the price of a forward on F's currency

Example 1

1. Borrow $\$_D 1\text{m}$ in D
2. Buy $\$_D 1\text{m}/E_0$ of F's currency (in $\$_F$)
3. Buy $\$_D 1\text{m}/E_0$ of F's risk-free
4. Sell a forward in F with price K_0 (in $\$_D$)

Example 1

5. In one year, we get

$$(1 + r_F) \cdot (\$_D 1\text{m}/E_0) \tag{12}$$

   in $\$_F$

6. We sell our F currency for K_0 and get

$$K_0 \cdot (1 + r_F) \cdot (\$_D 1\text{m}/E_0) \tag{13}$$

   in $\$_D$

7. We repay our loan

$$K_0 \cdot (1 + r_F) \cdot (\$_D 1\text{m}/E_0) - (1 + r_D) \cdot \$_D 1\text{m} \tag{14}$$

   in $\$_D$

Example 1

- You should <u>not</u> be able to make money from this trade
- It requires no money down
- We conclude that

$$1 + r_D \overset{\text{“should”}}{=} (1 + r_F) \cdot \frac{K_0}{E_0} \tag{15}$$

- **Warning:** the forward price on a D asset is $K_0 = S_0(1 + r_D)$. F's currency is <u>not</u> a D asset (it has its own rate r_F). The forward price on an F asset is its own thing.

Board Problem 1

Suppose that

1. the risk-free rate is $r_D = 5\%$ in the US
2. a euro sells for $E_0 = \$.05$
3. the forward price on pesos is

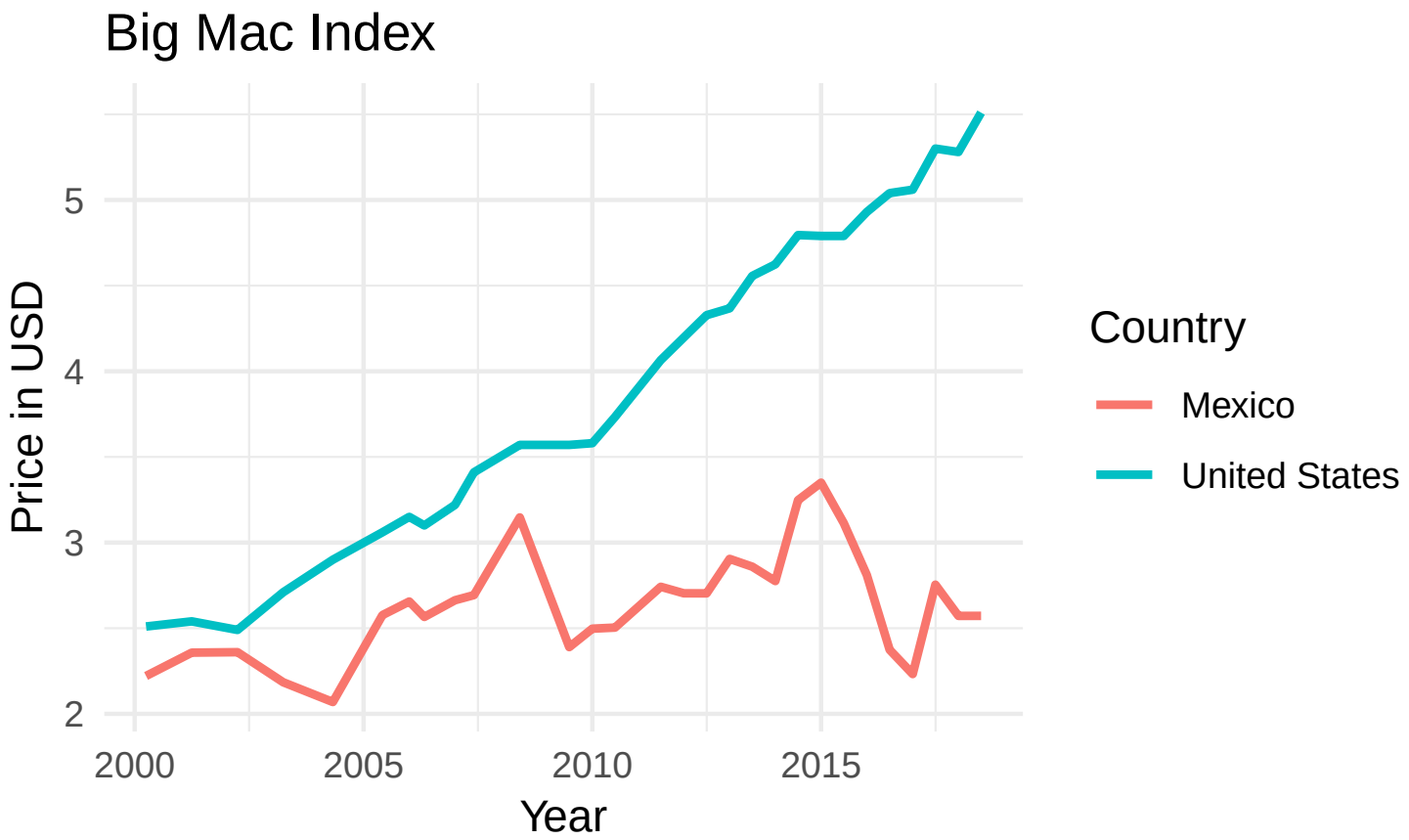
What should the risk-free rate be in the EU? What arbitrage opportunity is available to you if the risk-free rate in the EU is $r_F = 10\%$?

Example 2

- A taco in the US sells for USD$3
- A taco in Mexico sells for MEX$20
- A peso sells for $.05.
- What arbitrage opportunity is available to you?

1. Buy MEX$20 for USD$1
2. Buy 1 taco in Mexico for MEX$20
3. Sell 1 taco in the US for USD$3
4. You've made USD$2

The Economist's Big-Mac Index



†Courtesy of The Economist.